

(Heat Transfer)

Pre-Lab Plan
Unit Operations Lab 1
Date 1/20/26

Group 1, Tuesday, Section

Team member Names

Aaron Casas

Anna Cai

Hussein Rezk

Adil Feratovic

1. Experimental Objective (5 pts)

Using the stirred tank with water and steam for heat, the heat transfer and its coefficients, as well as the flow pattern, will be determined. Control systems, such as impeller spin direction and speed to adjust flow, as well as baffles to affect flow patterns, including turbulent and laminar flow.

2. Experimental

The stirred tank vessel contains many systematic units, such as an impeller, cooling coils, and a heat jacket, to reduce heat loss. A water piping stream and a steam piping stream are attached to the vessel to allow seamless entry to the tank. The water is pumped, then heat is exchanged, creating a closed loop through the tank vessel. Heat loss may occur at the top of the tank due to the lack of a heat jacket on the enclosures.

2.1 Apparatus (5 pts)



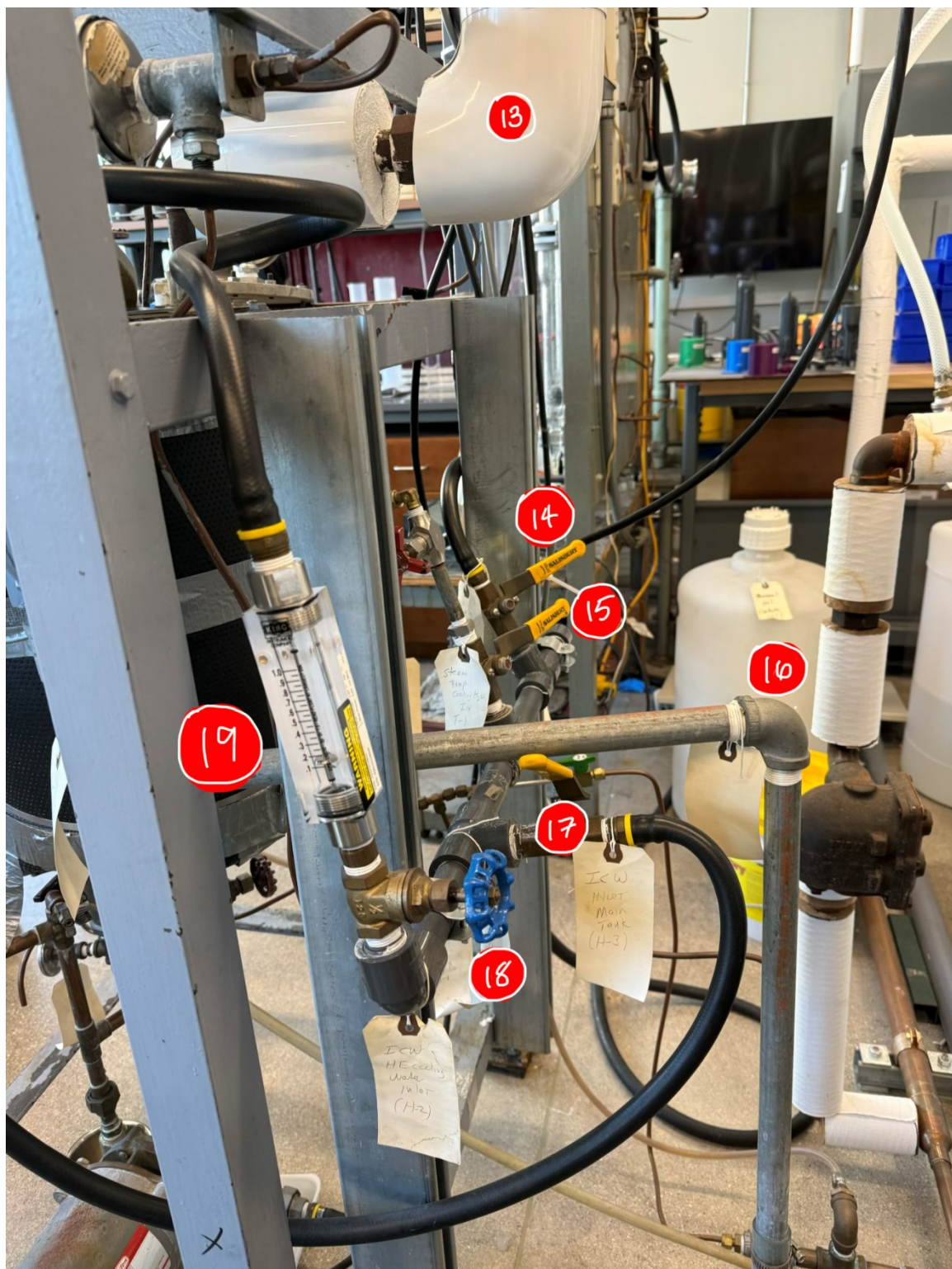




Table 1: List of Apparatus Components and Descriptions

Number	Component	Description
1	Recycled Water Rotameter	Shows the flowrate of recycled water from the heat exchanger
2	Temperature gauge	Measures the temperature of the inlet water to the heat exchanger from the tank
3	Temperature gauge	Measures the temperature of the inlet water to the heat exchanger to the tank
4	Jacketed tank	Steam enters the jacket to heat the system
5	Motor	Supplies power to the impeller
6	Temperature gauge	Measures the temperature of the stir tank
7	Valve	Controls the flow of the water to the inlet cooling coil
8	Cooling Coil Rotameter	Condenses the steam
9	Steam trap	Measures the temperature of the outlet cooling water from the steam condenser
10	Temperature gauge	Measures the temperature of the inlet cooling water into the steam condenser
11	Water line	Supplies water to the tank
12	Pump	Pumps water out of the tank
13	Steam Valve	Valve that allows steam to enter the system
14	Cooling Water Valve	Valve that allows cooling water to go to the coils
15	Condenser Valve	Valve that allows cooling water into the condenser
16	Heat exchanger Drainpipe	Allows there to be a flow through the heat exchanger
17	Tank Valve	Inlet to the main tank
18	Valve	Allows cold water to flow to the heat exchanger
19	Rotameter	Measures the flow rate of cooling water into the heat exchanger
20	Baffles	To increase the turbulence inside the tank
21	Impeller	Stirs the fluid inside the tank
22	Cooling coil	Cools the fluid inside the tank

2.2 Materials and Supplies (5 pts)

Table 2: List of Materials and Supplies

Material	Description
Water	Sourced from lab piping into the tank
Steam	Sourced from lab piping to adjust heat
Heat-Resistant Gloves	To ensure safe handling of baffles

	from the extreme heat produced by the steam
Thermometer/Temperature Gauge	Used to take the internal temperature of the tank.
Graduated Cylinder/Bucket	To measure the flow rate over time
Stopwatch/phone	To time the flow rate

2.3 Experimental Plan (10 pts)

2.4 Procedure:

Start Up Procedure

- Fill the vessel with water to around 2 inches from top and put lid
- Begin stirring with 300 RPM. Fluctuation in reading is normal. Take note of impeller rotation
- Start cooling water to V1 and V2, cooling coils
- Obtain 50% flowrate in rotameter by opening control valve
- Start cooling water to V11, external heat exchanger, and adjust flow to 0.5 LPM
- Open V5, condenser valve
- To prime the pump, close outlet valve (V9) in the pump. Open suction valve (V12) and bypass valve (V6). Turn pump on and allow it to run for a few seconds
- Open V4 and slowly open V9 to 100% flow, then close V6. Keep water flow at 100%, then reduce to 50%
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Experiment 1: Unbaffled Stirred Tank

- Slightly open V3 (1 – 1 ½ turns)
- Adjust recirculation/coil flow so that the tank temperature stays between 65 – 85 C
- After steady flow is reached, record:
 1. Flow rates of cooling water and recirculation flow
 2. T1 – T7
 3. Amount of steam condensate collected over time range

Experiment 2: Baffled Stirred Tank

- Same as experiment 1 but with baffles inserted into the system

Run Plan

Design: Comparison between baffled and unbaffled systems across three impeller speeds.

Replicate Data: 2 sets of replicate data will be collected

Independent Variables: Impeller Speed (3 levels), baffles, and tank temperature setpoint

Dependent Variables: Heat transfer rate to tank Q_{13} , overall coefficient U_i , and h_i

12 total runs

Tank temperature should be between 65 – 85 C

Uncertainty Reduction

- Verify thermometer, if needed
- Verify rotameter and flowmeters, if needed
- 2 sets of replicate data
- Confidence interval of 95%
- Goodness of fit will be measured by standard deviations and by looking at the R^2 values

Data Reduction and Theory Correlation

- Compute Q_{13} at steady state using the stirred tank energy balance
- Compute U_i from $Q_{13} = U_i A_i (t_1 - t_3)$
- Estimate h_0 using condensation correlation chart
- Calculate h_i from $U_i = (1/h_i + 1/h_0)^{-1}$

Project Safety Evaluation (10 pts)

Project Title: Heat Transfer

Date: 1/20/2026

	Yes/No	
Potential electrical hazards?	<u>yes</u>	If yes, identify conditions: The motor powering the mixer as well as the pump are both powered electrically which are both prone to water damage potentially causing water hazards.
Potential mechanical hazards?	<u>yes</u>	If yes, identify conditions: The main mechanical hazard in this experiment are the impellers as they do rotate potentially posing as a hazard of getting something stuck in the vessel during operation.
Potential pressure hazards?	<u>no</u>	If yes, identify conditions:
Potential temperature hazards?	<u>yes</u>	If yes, identify conditions: The steam pipe as well as the vessel can reach high temperatures that may burn the operator on contact.
Hazardous/toxic chemicals involved?	no	If yes, hazards involved:
Gloves required?	yes	If yes, specify type: Heat resistant gloves.
Other hazards		If yes, specify conditions: Another potential Hazard can be the overfilling of the vessel with water potentially causing leaks which may lead to water damage and slipping hazards.
MSDS reviewed by all team members?	yes	List MSDS sheets reviewed: (7732-18-5) Steam
<i>Process Parameters</i>	<i>Max Expected</i>	<i>List location and possible hazards and precautions</i>
Temperature (°C)	100 65-85	Within the steam piping the maximum temperature to expect is roughly 100 degrees Celsius. Within the vessel it can be expected to be within the range of 65 and 85 degrees Celsius once all the process variables (Cooling coil flowrate, recirculation flowrate in external heat exchanger, and the steam rate) are all in equilibrium.
Pressure (psia)		
<i>Safety Controls</i>	<i>yes/no</i>	<i>If yes, Provide description</i>
Valves	Yes	Valves give the operator full control of both cooling water and stem flowrates, allowing them to close in a moment of emergency.
Drain valves		Drain valves are also present in case there's a overflow of liquid within the system.
Rpm regulator		If the impeller poses as a hazard the operator has full control to turn of the impeller with the rpm selector moving it all the way down to zero.

I have read relevant background material for the Unit Operations Laboratory entitled:
“_____” and
understand the hazards associated with conducting this experiment. I have planned out my
experimental work in accordance to standards and acceptable safety practices and will conduct all
of my experimental work in a careful and safe manner. I will also be aware of my surroundings,
my group members, and other lab students, and will look out for their safety as well.

<u>Adil Feratovic</u>	<u>01/20/2026</u>
signature of team member	date

<u>Hussein Rezk</u>	<u>01/20/26</u>
signature of team member	date

<u>Aaron Casas</u>	<u>1-20-26</u>
signature of team member	date

<u>Anna Cai</u>	<u>1/20/26</u>
signature of team member	date